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Table 2: Differences between Tropic Movements and Nastic movements	
Tropic Movements	**Nastic movements**
1. Movement occurs due to unidirectional stimulus.	These movements occur due to a diffused stimulus.
2. The stimulus acts on protoplasm from one direction only.	The stimulus acts on the protoplasm from all sides.
3. The response is directly related to the direction of the stimulus.	The response has no relation to the direction of the stimulus but with organ.
4. These are movements of curvature caused by unilateral growth.	These are also the movement of curvature but they are caused by reversible turgor changes.
5. Tropic movements may be phototropic, geotropic, hydrotropic, thigmotropic, chemotropic, thermotropic or aerotropic.	Nastic movements may be seismonastic, photonastic or thermonastic

movement or tropism. There are seven types in tropic movements (Geotropic, Phototropic, Thigmotropic, Chemotropic, Hydrotropic, Thermotropic and Aerotropic)

1. Geotropism

The movements which take place in response to gravity stimulus are called **geotropic movements**. The primary roots growing down into soil are **positive geotropic.** Primary stems that grow away from soil (against gravity) are **negative geotropic**. Secondary roots growing at right angles to the force of gravity are **Diageotropic**. Secondary lateral roots which grow obliquely downwards are **Plagiogeotropic**. Lateral roots and branches which are not sensitive to gravitational stimulus are **Apogeotropic**.

2. Phototropism

The tropic movement taking place as a response to light stimulus is called **phototropism**. Some of the plant parts such as stems, branches, leaves and pedicels of flowers move towards the stimulus of light and are said to be **positively phototropic** while others such as roots and rhizoids which move away from the stimulus of light are called **negatively phototropic**.

b. Nastic Movements

When growth movements occur in response to an external stimulus which is not unidirectional but diffused, they are called nastic or paratonic movements of variation. Paratonic variation movements are determined by some external stimuli, light, temperature, chemicals and touch. They are:

1. Nyctinastic movement (or) sleep movement

The diurnal (change in day-night) movements of leaves and flowers of some species which take up sleep position at night are called **nyctinastic movements**. They are caused by relative changes in cell size on the opposite sides of the leaf base called **pulvinus**. The movements